

LABORATORY ASSIGNMENT

Course: Full Stack Web Development

Course Code: 23CM4121

Assignment Topic: JavaScript Inheritance Types (ES6)

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1. TITLE

Implementation of JavaScript ES6 Inheritance Types (Single, Multilevel, and Hierarchical)

2. AIM

To design and implement JavaScript ES6 class programs demonstrating Single Inheritance, Multilevel Inheritance, and Hierarchical Inheritance using the class and extends keywords.

3. OBJECTIVE

The objectives of this assignment are:

- To understand Object-Oriented Programming (OOP) concepts in modern JavaScript (ES6+).
- To implement Single Inheritance where a child class extends a single parent class.
- To implement Multilevel Inheritance where a derived class inherits from another derived class forming a chain.
- To implement Hierarchical Inheritance where multiple child classes inherit from a single parent class.

4. THEORY

Inheritance is a fundamental concept in Object-Oriented Programming that allows a class (child/subclass) to acquire properties and methods from another class (parent/superclass). ES6 introduced standard class and extends syntax for inheritance in JavaScript.

1. Single Inheritance: A child class inherits directly from one parent class.

2. Multilevel Inheritance: A class extends a parent class which itself extends another base class (Class A -> Class B -> Class C).

3. Hierarchical Inheritance: Multiple child classes extend the same base class (Class A -> Class B, Class A -> Class C).

5. PROGRAM IMPLEMENTATIONS

5.1 Single Inheritance (single.js)

// Single Inheritance Example in JavaScript (ES6)

```
class Animal {
  eat() {
    console.log("Animal eats food.");
  }
}

class Dog extends Animal {
  bark() {
    console.log("Dog barks: Woof! Woof!");
  }
}
```

```
    }  
  }  
  
  let d = new Dog();  
  d.eat();  
  d.bark();
```

5.2 Multilevel Inheritance (multilevel.js)

// Multilevel Inheritance Example in JavaScript (ES6)

```
class Vehicle {  
  startEngine() {  
    console.log("Vehicle engine started");  
  }  
}  
  
class Car extends Vehicle {  
  drive() {  
    console.log("Car is driving on the road");  
  }  
}  
  
class SportsCar extends Car {  
  turboBoost() {  
    console.log("SportsCar turbo boost activated!");  
  }  
}  
  
let sc = new SportsCar();  
sc.startEngine();  
sc.drive();  
sc.turboBoost();
```

5.3 Hierarchical Inheritance (hierarchical.js)

// Hierarchical Inheritance Example in JavaScript (ES6)

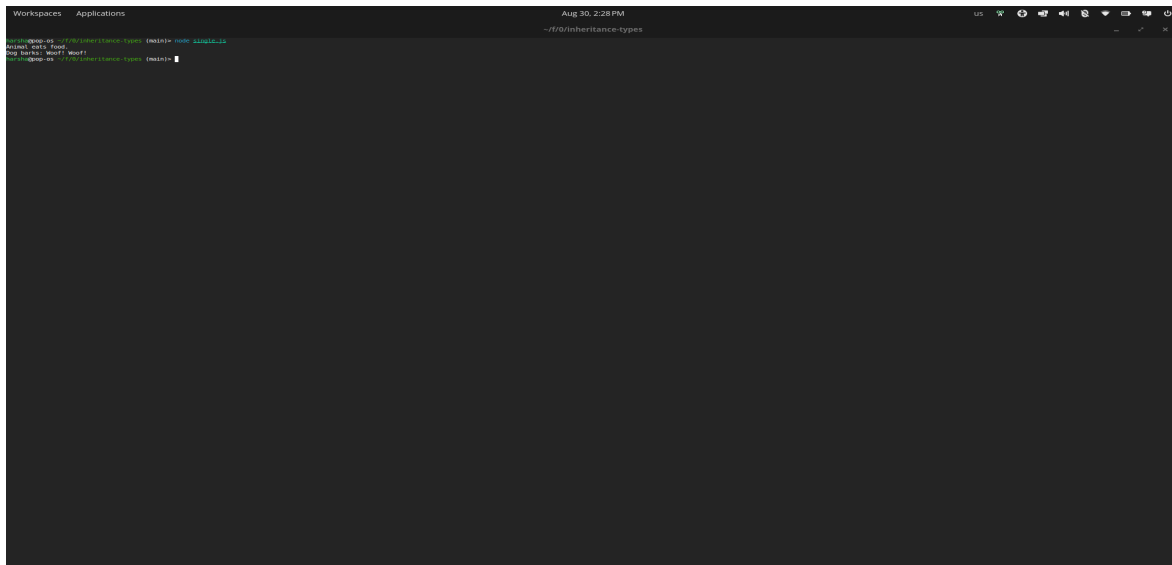
```
class Instrument {  
  play() {  
    console.log("Instrument can play");  
  }  
}  
  
class Guitar extends Instrument {  
  strum() {  
    console.log("Guitar is strumming");  
  }  
}  
  
class Piano extends Instrument {  
  press() {  
    console.log("Piano is playing");  
  }  
}  
  
let g = new Guitar();  
let p = new Piano();
```

```
g.play();  
g.strum();
```

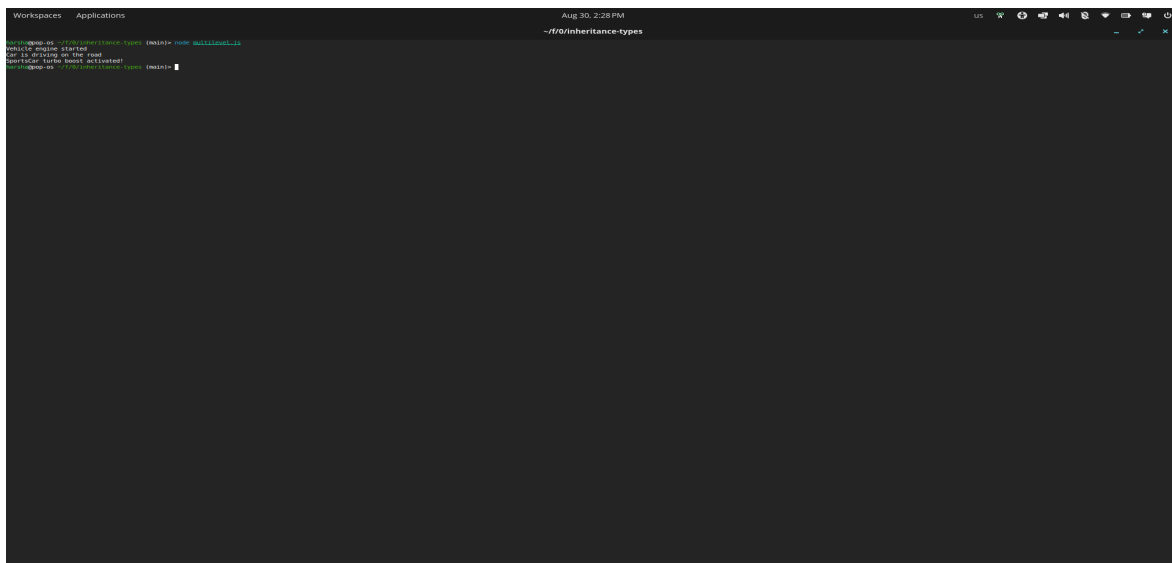
```
p.play();  
p.press();
```

6. EXECUTION OUTPUT SCREENSHOTS

Output 1: Single Inheritance Execution



Output 2: Multilevel Inheritance Execution



Output 3: Hierarchical Inheritance Execution

Thus, the three types of JavaScript inheritance (Single, Multilevel, and Hierarchical) were successfully implemented using ES6 classes and verified through Node.js execution.